MATH*6210 A1

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Toy model of glycemic control:

$$u = \alpha v + \beta \int_{-\infty}^{t} \lambda e^{-\lambda(t-t')} v(t') dt' \tag{1}$$

$$\dot{v} = -\gamma v - \delta u(v + \bar{v}) + F(t) \tag{2}$$

1. Starting from the model with one integral and one differential equation, derive an ODE with 2 DOF

Using Leibniz's rule:

$$\dot{u}(t) = \alpha \dot{v} + \beta \left[\lambda e^{-\lambda(t-t)} v(t) \cdot (1) + \int_{-\infty}^{t} \lambda \left(-\lambda e^{-\lambda(t-t')} \right) v(t') \, dt' \right]$$

Setting $e^0 = 1$ and factoring out $-\lambda$ from the integral:

$$\dot{u} = \alpha \dot{v} + \beta\lambda v - \beta\lambda \int_{-\infty}^{t} \lambda e^{-\lambda(t-t')} v(t') \, dt'$$

$$\text{Sub in: } \beta \int_{-\infty}^{t} \lambda e^{-\lambda(t-t')} v(t') \, dt' = u - \alpha v \quad \text{from equation (1)}$$

$$\begin{aligned} \dot{u} &= \alpha \dot{v} + \beta\lambda v - \lambda(u - \alpha v) \\ &= \alpha \dot{v} + \beta\lambda v - \lambda u + \alpha\lambda v \\ \dot{u} + \lambda u &= \alpha \dot{v} + \lambda(\alpha + \beta) v \end{aligned}$$

Substituting $\dot{v} = -\gamma v - \delta u(v + \bar{v}) + F(t)$ from equation (2):

$$\dot{u} = -\lambda u + \lambda(\alpha + \beta) v + \alpha \left(-\gamma v - \delta u(v + \bar{v}) + F(t) \right) \tag{3}$$

$$\dot{v} = -\gamma v - \delta u(v + \bar{v}) + F(t) \tag{4}$$

Final form:

$$\dot{u} = -\lambda u + \left[\lambda(\alpha + \beta) - \alpha\gamma \right] v - \alpha\delta u(v + \bar{v}) + \alpha F(t) \tag{5}$$

$$\dot{v} = -\gamma v - \delta u(v + \bar{v}) + F(t) \tag{6}$$

2. Scale the dependent variables and time to reduce the number of parameters. Follow the lecture and fill the gaps.

Let the scales be defined as:

$$v = \bar{v} v', \quad u = \tilde{u} u', \quad t = \tilde{t} t'$$

Recall equations (5) and (6). Substituting into the v equation:

$$\begin{aligned} \frac{1}{\tilde{t}} \bar{v} \frac{dv'}{dt'} &= -\gamma(\bar{v} v') - \delta(\tilde{u} u')(\bar{v} v' + \bar{v}) + F \\ \implies \frac{dv'}{dt'} &= -(\gamma \tilde{t}) v' - (\delta \tilde{t} \tilde{u}) u'(v' + 1) + \frac{\tilde{t}}{\bar{v}} F \quad \text{let } G = \frac{\tilde{t}}{\bar{v}} F \end{aligned}$$

Setting coefficients to 1:

$$\gamma\tilde{t} = 1 \implies \tilde{t} = \frac{1}{\gamma}$$

$$\delta\tilde{t}\tilde{u} = 1 \implies \tilde{u} = \frac{1}{\delta\tilde{t}} = \frac{\gamma}{\delta}$$

$$\implies \underbrace{\frac{dv'}{dt'} = -v' - u'(v'+1) + G}_{\text{call this } f'(u',v') \text{ (now dimensionless)}}$$

Substituting into the u equation:

$$\frac{1}{\tilde{t}}\tilde{u}\frac{du'}{dt'} = -\lambda\tilde{u}u' + \lambda(\alpha+\beta)\bar{v}v' + \alpha f$$

$$\implies \frac{du'}{dt'} = -(\lambda\tilde{t})u' + \left[\lambda\bar{v}(\alpha+\beta)\left(\frac{\tilde{t}}{\tilde{u}}\right)\right]v' + \left(\alpha\frac{\tilde{t}}{\tilde{u}}\right)f$$

Using $\tilde{t} = \frac{1}{\gamma}$ and $\tilde{u} = \frac{\gamma}{\delta}$:

$$\implies \frac{du'}{dt'} = -\left(\frac{\lambda}{\gamma}\right)u' + \left[\left(\frac{\lambda\bar{v}}{\gamma}\cdot\frac{\delta}{\gamma}\right)(\alpha+\beta)\right]v' + \left(\alpha\frac{\delta}{\gamma^2}\right)f'(u',v')$$

Let $a = \frac{\lambda}{\gamma}$, $\quad b = \frac{\lambda\bar{v}\delta}{\gamma^2}(\alpha+\beta)$, $\quad c = \frac{\alpha\delta}{\gamma^2}$

$$\implies \frac{du'}{dt'} = -au' + bv' + cf'(u',v')$$

3. Find the equilibria if any

Drop the primes:

$$-v - u(v+1) + G = 0$$
$$-au + bv = 0 \quad \text{(since } f(u,v) = 0 \text{ from the above equation)}$$

From the second equation:

$$-au + bv = 0 \iff u = \frac{b}{a}v$$

Substituting this into the first equation:

$$-v - u(v+1) + G = 0$$
$$\iff -v - \frac{b}{a}v(v+1) + G = 0$$
$$\iff -\frac{b}{a}v^2 - \frac{b}{a}v - v + G = 0$$
$$\iff \frac{b}{a}v^2 + \left(1 + \frac{b}{a}\right)v - G = 0$$
$$\iff v^2 + v + \frac{a}{b}v - \frac{a}{b}G = 0$$
$$\iff v^2 + \left(1 + \frac{a}{b}\right)v - \frac{a}{b}G = 0$$

Using the quadratic formula to solve for the equilibrium points v^*:

$$v^* = -\frac{1}{2}\left(1 + \frac{a}{b}\right) \pm \frac{1}{2}\sqrt{\left(1 + \frac{a}{b}\right)^2 + \frac{4aG}{b}}$$

Analysis of Solutions:

For $G > G_{\text{SN}}$, the system has two distinct equilibrium solutions corresponding to upper and lower branches. One is a stable node and the other is a saddle point. In contrast, exactly one equilibrium solution occurs at the saddle-node bifurcation point where the term under the square root vanishes, giving the critical input parameter $G_{\text{SN}} = -\left(1 + \frac{a}{b}\right)^2 \frac{b}{4a}$.

4. Exploring Controllability via Python script

Looking at the eigenvalues for this system gives us quantitative confirmation of local stability. For the set of baseline parameters ($p_0 = 0.1, p_1 = 1.0, p_2 = 1.0$) at a parameter value of $G = 2.0$, linearization around the upper equilibrium point (eq_0) gives us a Jacobian matrix with strictly negative eigenvalues ($\lambda_1 = -0.440, \lambda_2 = -3.091$), confirming its asymptotically stable node. In contrast, at the lower equilibrium point (eq_1) consistently yields real eigenvalues of opposite signs ($\lambda_1 = -0.118, \lambda_2 = 11.549$) with a negative determinant ($\det(J) < 0$), verifying that its an unstable saddle point. In general, the existence and character of these states remain strictly governed by the sign of the system discriminant D. Extending this analysis to different values of G reveals distinct operational regimes: increasing G widens the separation between equilibria and expands the safe basin of attraction, whereas decreasing G toward the bifurcation limit compresses the controllable phase space and heightens vulnerability to control failure.

Building upon this framework, the global phase portrait illustrates the broader system flow through vector field streamlines and explicit nullclines. The curves where $\frac{du}{dt} = 0$ (blue) and $\frac{dv}{dt} = 0$ (orange) intersect precisely at the equilibrium coordinates, providing graphical confirmation of the analytical solutions. Tracing the stable manifold of the saddle point maps the global basin of attraction and the precise separatrix boundary. Initial conditions lying on either side of this invariant curve dictate system fate: states within the basin successfully converge to the stable node, whereas those crossing the threshold trigger runaway divergence and safety interventions ($v < -100$), marking a total loss of glucose regulation.

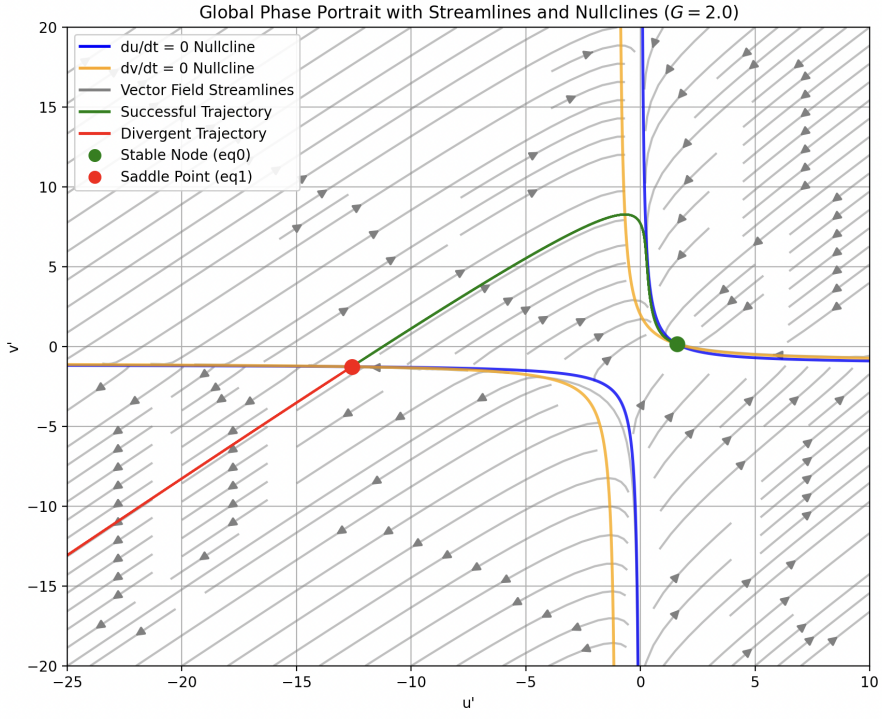


Figure 1: Global phase portrait for $G = 2.0$, featuring vector field streamlines, nullclines ($\frac{du}{dt} = 0$ in blue, $\frac{dv}{dt} = 0$ in orange), the stable node (eq_0), and the saddle point (eq_1).

These structural boundaries are tightly linked to parameter variations. Decreasing the input parameter G toward its critical threshold G_{SN} forces the stable node and saddle point closer together. At the bifurcation point, their eigenvalues collide and yield a zero eigenvalue, destroying the equilibria. Geometrically, this contraction severely shrinks the basin of attraction, heightening the system's sensitivity to perturbations.